॥ श्रीगणेशाय नमः ॥

Profit and Loss

Profit % = $\frac{\text{profit}}{CP} \times 100 = \frac{SP - CP}{CP} \times 100 = \left(\frac{SP}{CP} - 1\right) 100$ $\Rightarrow \frac{\text{profit \%}}{100} + 1 = \frac{SP}{CP}$

Loss % = $\frac{\text{loss}}{CP} \times 100 = \frac{CP - SP}{SP} \times 100$

Q. A buys a watch for rs. 350 and sells it for rs. 392. What's the %age of profit?

→ $\frac{392-350}{350} \times 100 = 12\%$

Q. B buys a bicycle for rs. 5200 and spends rs.800 on it's repairs. He sells it at rs. 5500. Find the %age of profit/loss?

→ Loss = $6000 - 5500 = 500$

Loss % = $\frac{500}{6000} \times 100 = 8.33$ 8.33 % loss

Q. A man buys 10 articles at rs.8 and sells them for rs. 1.25/article. What's the %age of profit?

→ $\frac{1.25 - 0.8}{0.8} \times 100$

Q. B sold his scooter for rs. 10,500 at a gain of 5%. Find the CP.

→ $\frac{10500 - x}{x} \times 100 = 5$

$10500000 - 100x = 5x$

$x = ₹10,000$

OR $SP = \frac{105}{100} CP$

$10500 = \frac{105}{100} CP$

$CP = ₹10,000$

Q. A watch is sold for rs. 880 at a loss of 20%. What's the CP?

→ 80 % CP = SP

$CP = \frac{880}{80} \times 100 = 1,100$

Q. A man bought pencils at the rate of 6 pencils for rs. 4 and sold them at the rate of 4 pencils for rs.6. What's the gain %age?

→ CP = $\frac{4}{6}$ per pencil SP = $\frac{6}{4}$ per pencil

$\text{gain} = \frac{\frac{6}{4} - \frac{4}{6}}{\frac{4}{6}} = \frac{5}{6} \times \frac{6}{4} = \frac{5}{4} = \frac{5}{4} \times 100\% = 125\%$

Q. If eggs are bought at rs.10/12 eggs and sold by rs.12/10 eggs, what's the gain or loss %?

→ CP: $\frac{10}{12}$ SP: $\frac{12}{10}$

$\text{gain \%} = \frac{\frac{12}{10} - \frac{10}{12}}{\frac{10}{12}} \times 100 = \frac{44}{120} \times \frac{12}{10} \times 100 = 44\%$ gain

Q. 4 lemons were bought at rs.3. 5 lemons were sold at rs.4. What's the %age profit?

→ $\frac{3}{4}$ $\frac{4}{5}$

$\frac{\frac{4}{5} - \frac{3}{4}}{\frac{3}{4}} \times 100 = \frac{1}{20} \times \frac{4}{3} \times 100 = 6.66\%$ gain

Q. Ramesh bought 2 horses for rs.1200 each. He sells 1 at 10% profit and the other at 10% loss. What's the overall profit/loss?

→ 1st horse SP = $\frac{110}{100} \cdot 1200$

2nd horse SP = $\frac{90}{100} \times 1200$

Total CP = $1200 \times 2 = 2400$

Total SP = $1200\left(\frac{110+90}{100}\right) = 12 \cdot 200 = 2400$

Profit % = $\frac{2400-2400}{2400} \times 100 = 0\%$

Q. Ramesh sold 2 horses for rs.1200 each. He sells 1 at 10% profit and the other at 10% loss. What's the overall profit/loss?

→ 2.5%

Q. Ramesh sold 2 horses for rs.12000 each. He sells 1 at 20% profit and the other at 20% loss. What's the overall profit/loss?

→ 1: $S_1 = \frac{120}{100} C_1$

$\frac{12000 \times 100}{120} = C_1 = 10{,}000$

2: $S_2 = \frac{80}{100} C_2$

$\frac{\overset{3}{\cancel{12000}} \times 100}{\underset{2}{\cancel{240}}} = C_2 = \frac{30000}{2} = 15{,}000$

overall CP = 25,000

overall SP = 24,000

loss % = $\frac{1000}{25000} \times 100 = 4\%$

If SP of 2 articles is same
and we get x% profit and x% loss resp.,
we get loss % = $\frac{x^2}{100}$

$\frac{20 \cdot 20}{100} = \underline{4}$

Q. 2 bicycles were sold for rs. 3990 each, gaining 5% on one and losing 5% on the other. Gain/loss % on the entire transaction?

→ $\frac{5 \cdot 5}{100} = 0.25\%$ loss

Q. A sold 2 books for rs.12,000 each, one at 20% profit and the other at 4% loss. Total profit/loss?

→ $12000 = \frac{120}{100} C_1$ $\quad 12000 = \frac{96}{100} C_2$

$C_1 = 10000$ $\quad C_2 = 12500$

CP = 22500

SP = 24,000

gain % = $\frac{24000-22500}{22500} \times 100 = 6.66\%$

Q. The cost of 2 watches together is rs. 840. If by selling them at a profit of 16% and a loss of 12% resp, we get no profit, no loss. CP of each watch?

→ $C_1 = x$, $C_2 = 840-x$

$S_1 = \frac{116}{100} x$ $\quad S_2 = \frac{88}{100}(840-x)$

$840 = \frac{116}{100} x + \frac{88}{100}(840-x)$

$x = \underline{360}$ $\quad 840-x = \underline{480}$

Q. A shopkeeper buys 2 fans at rs. 3120. He sells one with 36% profit and the other at 15% loss. Each fan was sold for the same price. What is the CP of each fan?

→ C_1: x, C_2: $3120-x$

$\frac{136}{100} x = \frac{85}{100}(3120-x)$

$136x = 265200 - 85x$

$221x = 265200$

$x = 1200$ $\quad 3120-x = 1920$

Consecutive sales

Q. Mohit sells a bicycle to Rohit at 10% profit, who further sells it to Jyoti at 5% profit. If Jyoti pays rs.462 to Rohit, what's the CP for Mohit?

→ let CP for mohit = x

SP = $\frac{110}{100}x$ = CP for Rohit

SP for Rohit = $\frac{105}{100} \times \frac{110}{100} x$ = CP for Jyoti = 462

$x = \frac{46200}{110 \times 105} = ₹400$

Q. A sells a book to B at 17% gain. B sells it to C at 25% loss. C pays rs. 1053 to B. CP for A?

→ $1053 = \frac{75}{100} \cdot \frac{117}{100} x$

$x = ₹1200$

Q. A sells an article to B at a 20% gain, who sells it to C at a 10% gain, who then sells it to D at 12.5% gain. D pays rs. 29.70, what did it cost to A?

→ $29.7 = \frac{112.5}{100} \times \frac{110}{100} \times \frac{120}{100} x$

$x = 20$

Discounts

NOTE: Discount is applied to the MRP / marked price.

Discount = MRP − SP

Discount % = $\frac{MP - SP}{MP} \times 100$

Q. A book is marked rs. 200. Say successive discounts are given: 10%, then 5% on the new price; Find the discount %age.

→ $D_1 = \frac{10}{100} \times 200 = 20$

price = 180

$D_2 = \frac{5}{100} \times 180 = 9$

price = 171

Discount % = $\frac{200 - 171}{200} \times 100 = 14.5\%$

OR $SP = 200 \times \frac{90}{100} \times \frac{95}{100} = 171$

Q. MP = rs.500. Sold at successive discounts: 20%, 10%. SP=?

→ $SP = 500 \times \frac{80}{100} \times \frac{90}{100} = ₹360$

combined discount = $20 + 10 - \frac{20 \times 10}{100} = 28\%$

Q. MP = rs.240. Sold at successive discounts: 10%, 5%. SP=?

→ $240 \cdot \frac{90}{100} \cdot \frac{95}{100} = 205.20$

Q. A single discount equivalent to successive discounts of 10%, 20%, 25% is ___.

→ $\frac{90}{100} \cdot \frac{80}{100} \cdot \frac{75}{100} = \frac{9}{10} \cdot \frac{4}{5} \cdot \frac{3}{4} = \frac{27}{50} = \frac{54}{100}$

ans = 46%

Discount = 46%

Q. MP = rs. 800. You get 2 successive discounts and sells the watch at rs. 612. First discount is 10%, what's the 2nd?

→ $612 = 800 \times \frac{90}{100} \times \frac{x}{100}$

$x = 85$

15%

Q. Price = rs. 160. Customer buys it for rs. 122.40 after 2 successive discounts. 2nd discount is 15%, what's the 1st?

→ $122.4 = 160 \times \frac{85}{100} \times \frac{x}{100}$

$x = 90$

2nd discount = 10%

Q. A book was sold for rs.6300 after discounts 12.5% and 10%. Find the marked price.

→ $6300 = x \cdot \frac{87.5}{100} \cdot \frac{90}{100}$

$x = ₹8000$